Progress meeting

Optimisation-driven quantum circuit reduction

Reproduction, exact Clifford ground truth, and a general hardware-aware optimiser

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Study of B. Rosenhahn, T. J. Osborne, C. Hirche, *Optimization driven quantum circuit reduction*, New J. Phys. **27**, 104509 (2025), arXiv:2502.14715

The story in one slide

What I was asked to do

- Understand the paper, re-run every figure and table.
- Find its weaknesses.
- Extend it.

What I found

- It reproduces, but with qualifications.
- Every gate it uses is **Clifford**, so exact optima are computable.
- Against them, the method is **3–4.5**$\times$ from optimal.

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What I built

- Theory: when local rewriting is globally optimal; parity; what random benchmarks measure.
- A **general optimiser**: no Clifford assumption, real native gates, connectivity, calibrated noise, exact / state-aware / approximate semantics.

Where it stands

- Proven optimum on 35/40 of the paper's 3-qubit circuits.
- On top of Qiskit L3 on 58 real algorithms: never more 2q gates, $-13\,\%$ gates on IBM Heron.
- Extension paper drafted (12 pages); every output verified.

Roadmap

1. **The paper**: database-driven local rewriting
2. **Reproduction**: what holds, what needs qualifying
3. **Ground truth**: every gate is Clifford $\Rightarrow$ exact optima
4. **Theory**: exactness, parity, mixing, lower bounds
5. **A general optimiser**: no Clifford crutch, real hardware
6. **Results**: paper benchmarks, standard suites, noisy simulation
7. **Wrap-up**: lessons, caveats, critique, next steps

Part I

The paper

Rosenhahn, Osborne & Hirche: shorten circuits by replacing short windows with shorter equivalents from a precomputed database.

Why shorten circuits at all?

- On NISQ hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  with two-qubit errors $\sim 10^{-3}$–10^{-2}, 10–100$\times$ single-qubit ones.

- Compiling to a device's **native alphabet** leaves redundancy: one textbook gate becomes several native ones, and neighbours stop cancelling.

- Goal: an equivalent circuit with fewer (or cheaper) native gates.

Figure: arXiv:2502.14715, Fig. 1

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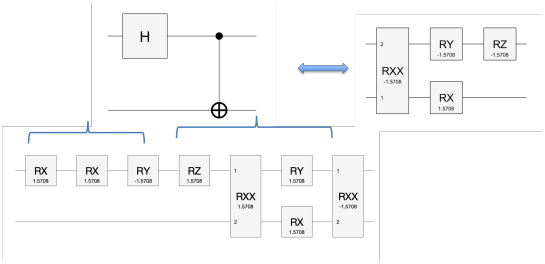
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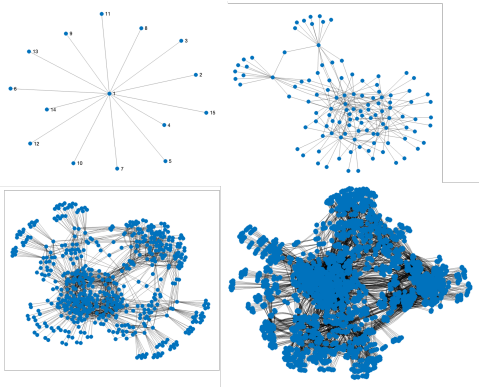


The paper's motivating example: H + CNOT compiled to the ion-trap alphabet (8 gates) and the shortest equivalent native word (4 gates). We confirmed by exhaustive search that no 3-gate version exists.

Figure: arXiv:2502.14715, Fig. 1

Step 1: the compute graph, a database of shortest circuits

- Fix a **discrete alphabet** on 2–3 qubits:
  - ion trap: $R_x, R_y, R_z(\pm\frac{\pi}{2})$, $R_{XX}(\pm\frac{\pi}{2})$
  - IBM-style: $R_x, R_z(\pm\frac{\pi}{2})$, CZ
- Breadth-first search over words: each node is a unitary (up to global phase), stored with its *shortest* word.
- Depth $d =$ maximal word length enumerated.



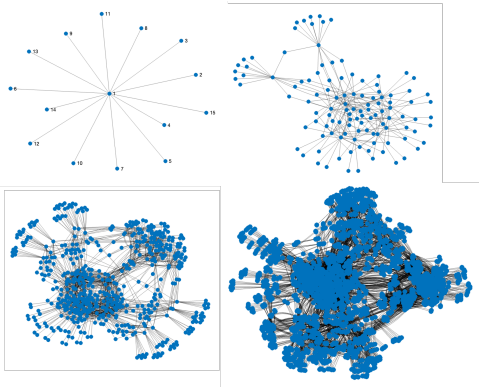
Compute graphs of growing depth (paper, Fig. 2).

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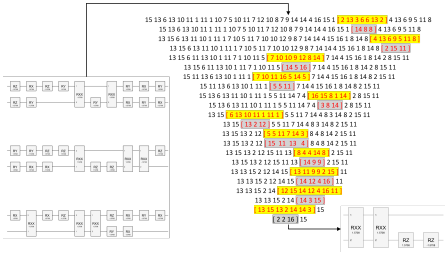
depth	1	2	3	4	5
2q, 14 ops	15	114	584	2,024	4,512
3q, 24 ops	25	337	3,215	23,622	137,572

Table I node counts: our rebuild matches the paper exactly.



Compute graphs of growing depth (paper, Fig. 2).

Step 2: slide a window, replace, repeat

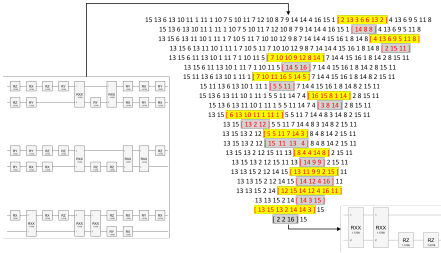


Circuit as a word of tokens; windows are replaced by shorter database words (paper, Fig. 3).

- Pick a window of 2–8 gates on ≤ 3 wires.
- Look its unitary up; if a shorter word exists, substitute it.
- Shuffle commuting gates to expose new windows; repeat until no progress.

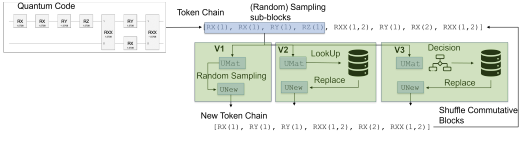
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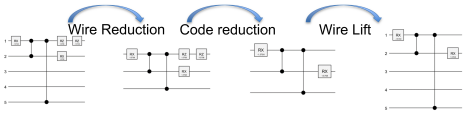
- Pick a window of 2–8 gates on ≤ 3 wires.
- Look its unitary up; if a shorter word exists, substitute it.
- Shuffle commuting gates to expose new windows; repeat until no progress.



- **V1**: random sampling for a shorter equivalent word.
- **V2**: database lookup.
- **V3**: a random forest first predicts whether a window is reducible; only then look up.

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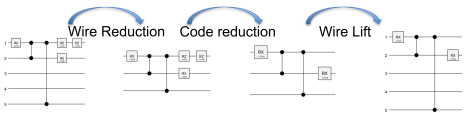
Step 3: scaling up, and what the paper claims



Wire reduction / code reduction / wire lift: a 3-qubit database is applied inside an n-qubit circuit.

Takeaway. A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

Step 3: scaling up, and what the paper claims



Wire reduction / code reduction / wire lift: a 3-qubit database is applied inside an n-qubit circuit.

Claims

benchmark (random native circuits)	paper	Qiskit (theirs)
T. III: ion, 3q, 300 gates	45	176
T. IV: IBM-style, 3q, 300 gates	72	153
T. V: IBM-style, 15q, 500 gates	293	313
journal: 4q, 300 gates (ion / IBM)	111 / 107	–
T. II time (s): V1 / V2 / V3	199 / 55 / 38	

- Fig. 9: reduced circuits give better results on IBM Brisbane and Kyiv.

Takeaway. A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

Part II

Reproduction

Re-running every figure and table with a Python port of the authors' own MATLAB code.

What I started from, and what I built

Inputs

- arXiv PDF + LaTeX source, journal (NJP) PDF.
- Official code `QCOptimDemo.zip`: linked only in the journal's ref. [38]; **V2 only**, MATLAB.
- Unspecified in the paper: database for Tables III–V, Table II protocol, forest details, transpiler level, Qiskit version.

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Built

- `qcr/`: line-by-line Python port (graph builder, V1/V2/V3, wire reduction), with tests.
- An independent exact engine (Clifford tableaux) as a cross-check.
- Every output checked equivalent to its input (unitary error $\sim 10^{-12}$).
- Scripts + raw results for each table and figure; reproduction report (PDF).

Verdict at a glance

Claim	Status	Detail
Table I (compute graphs)	**exact**	identical node and edge counts
Figs. 1, 4, 5 (examples)	**yes**	same end points; Fig. 4 needs a different gate pool (parity, Part IV)
Tables III/IV, journal 4q	**qualified**	only with the 24-operator, depth-5 database (unstated)
Table II: V1 $>$ V2 $>$ V3	**qualified**	V3 wins only with a tuned forest threshold
Table V (15 qubits)	**no longer**	ties a current Qiskit (2.5)
Fig. 9 (hardware)	**qualified**	real at Qiskit levels 0/1, disappears at level 3

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Takeaway. The headline claim (beats Qiskit on few-qubit random circuits) holds; the details around it need qualifying.

Table II: V1 vs. V2 vs. V3

variant	mean (s)	median (s)	reached	lookups
V1 random sampling	254.0	134.2	73	51,350
V2 database	46.4	6.5	96	7,904
V3 forest, $t = 0.5$	171.8	16.9	79	915
V3 forest, $t = 0.2$	33.7	5.3	98	1,216

100 random 5-qubit ion circuits, length 100 $\rightarrow$ target 60, cap 600 s. Paper: 199 / 55 / 38 s.

Fig. 7 reproduction: 100 random 5-qubit circuits, length 100 → 60

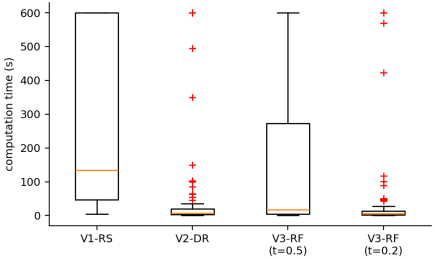
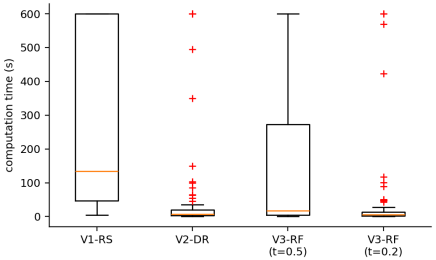


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- V1 ≫ V2 holds: the database is a real speed-up.
- V3 depends entirely on the forest's **recall**: on held-out windows it flags only **30 %** of reducible ones at the default threshold 0.5 (92 % at 0.2).
- At 0.5 it discards most reductions and is *slower* than V2. The paper reports neither features, forest size, threshold nor accuracy.

Takeaway. The "learning" variant only helps after tuning a threshold the paper does not mention, and it saves time on a lookup that need not be slow.

Tables III–V and the journal statistics: comparison with Qiskit

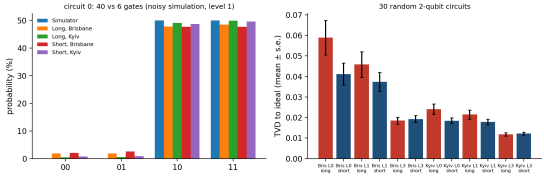
benchmark	input	Qiskit (paper)	Qiskit 2.5 L3	paper	our faithful port
T. III ion, 3q (20 circuits)	300	176	127.2	45	**47.1**
T. IV IBM-style, 3q (20)	300	153	104.0	72	**36.8**
journal ion, 4q (30)	300	–	164.9	111	**78.8**
journal IBM-style, 4q (30)	300	–	140.9	107	**76.0**
T. V IBM-style, 15q (10)	500	313	332.1	293	**335.9**

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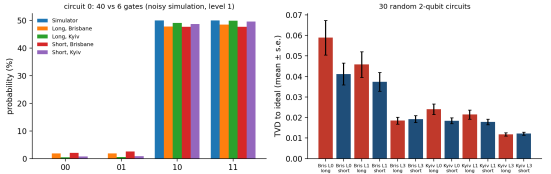
- Few qubits: the method clearly beats Qiskit, as claimed; our port is even better than published.
- But only with the larger database (24 operators, depth 5), which the paper does not state.
- Qiskit 2.5 is much stronger than the paper's Qiskit at 3 qubits (127 vs. 176; 104 vs. 153).
- 15 qubits: our port reaches 335.9 (paper 293) and only ties Qiskit L3 (332.1, same CZ count). Random 15-qubit gate pairs rarely share wires, so 3-wire windows find little that Qiskit misses.

Fig. 9: the hardware experiment, in a noisy simulator



TVD to ideal (30 circuits)	level 0	level 1	level 3
Brisbane: long → reduced	0.059 → 0.041	0.046 → 0.037	0.018 vs. 0.019
Kyiv: long → reduced	0.024 → 0.018	0.021 → 0.018	0.012 vs. 0.012

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- Levels 0/1: the reduced circuit is closer to ideal (22/30 and 20/30 circuits), as in the paper.
- Level 3: Qiskit re-synthesises every 2-qubit block, both versions compile to ≈ 1.1 CZ, the effect **vanishes**.
- Plain Qiskit L3 on the *long* circuit beats the reduced circuit at level 1 (0.018 vs. 0.037).
- Paper: one circuit, one run per chip, no error bars, transpiler level not stated.

Bugs and unstated choices found along the way

In the released code

1. `simplereduce` never tests the last adjacent pair, and skips the pair after a deletion: the final circuit in the paper's Fig. 4 still contains an inverse pair (true optimum 2, shown 4).
2. The database excludes the identity, so identity blocks are never deleted by lookup.
3. Three tolerances: 10^{-5} (paper), 10^{-3} (optimiser), 10^{-4} (graph builder).

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In the paper

4. Gate pool: Table I implies 14/24 operators, the demo uses 13/21.
5. Database for Tables III–V unstated; only the larger one reproduces Table III.
6. Table II: qubit count and target length unstated; forest unspecified.
7. Hardware: text says RX, RY, CX; tables say RX, RZ, CZ; transpiler level and shots unstated.

Part III

Exact ground truth

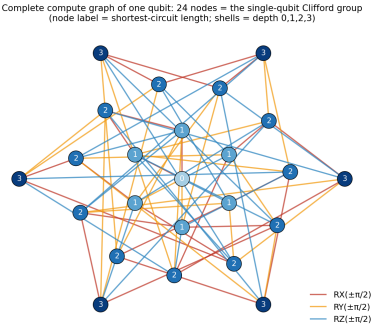
The observation the paper misses: every gate it uses is Clifford, so the search space is a finite group we can enumerate completely.

Key observation: every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group** $\mathcal{C}_n$ (mod phase), of order

$$2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$$

  $n=1$: 24, $n=2$: 11,520, $n=3$: **92,897,280**.
- The paper's "compute graph" is a ball in the **Cayley graph** of this group.



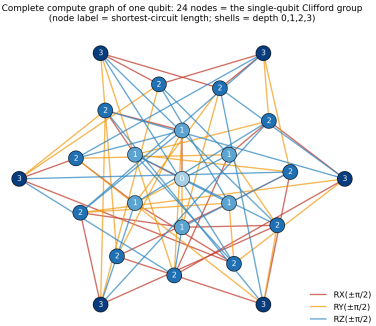
One qubit: the complete compute graph has exactly 24 nodes and diameter 3.

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Takeaway. Optimality is no longer a vague question: each circuit has an exact shortest length, computable by search over one finite graph.

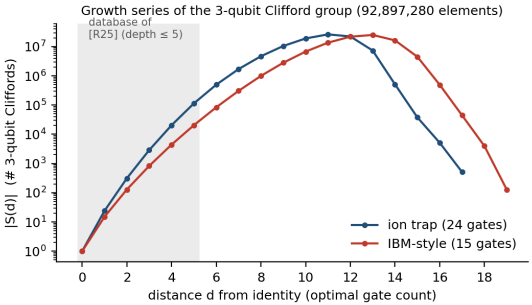
Enumerating all 92.9 million 3-qubit Cliffords

- A Clifford mod phase = its signed **tableau**; for $n \leq 3$ it packs into one 64-bit word.
- Gates act row-wise through lookup tables: each BFS level is a vectorised gather. Exact membership, no tolerance.
- Full BFS: 25 min, ≈6 GB, per alphabet.

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alphabet	n	diameter	mean optimal length
ion trap	2	10	5.89
IBM-style	2	11	6.92
ion trap	3	**17**	**10.72**
IBM-style	3	**19**	**12.34**



The paper's depth-5 database (shaded) covers 0.15 % of the group; almost all elements sit at distance 10–14.

How far from optimal is the paper's method?

3 qubits, 300 random gates, 20 circuits	**proven optimum**	paper's method (our port)	Qiskit 2.5 L3	paper (1 circuit)
ion trap	**10.4**	47.1	127.2	45
IBM-style	**12.2**	36.8	104.0	72

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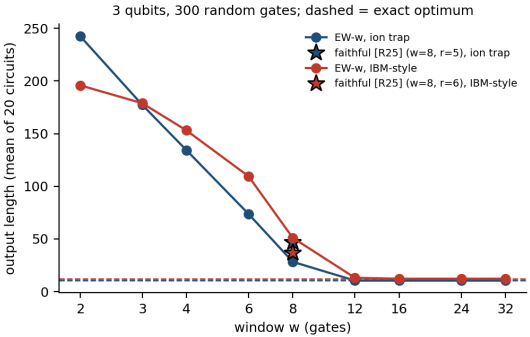
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The gap

The method's output is **4.5**× (ion trap) and **3.0**× (IBM-style) the optimum, on its own benchmark. Its showcase Fig. 4 output is 2× optimal.

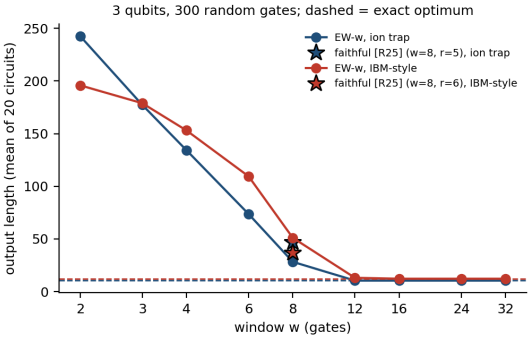
- Our port matches the paper's own single-circuit number (47.1 vs. 45), so this is the method, not the implementation.
- Beating Qiskit by 2.7× looked impressive; another 4.5× was available.

Exact windows close the gap



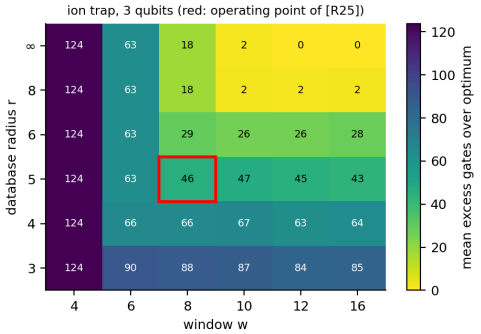
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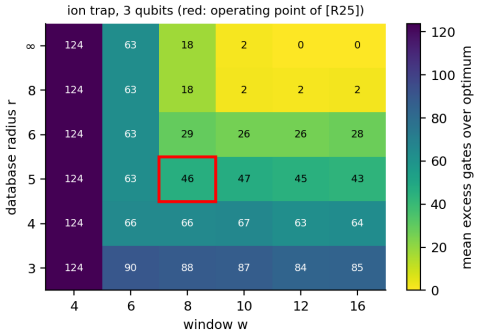
- **Exact-window optimiser** EW-w: replace every window of $\leq w$ gates on ≤ 3 wires by its *proven* optimal word (table lookup).
- Sharp transition: $w \geq 12$ (ion) / $w \geq 16$ (IBM-style) reaches the **proven optimum on all 40 circuits**.
- 0.1–0.2 s per circuit, vs. 51 s / 17 s for the paper's method.
- Why so sharp? Once the circuit is shorter than the window, one more sweep is exact (the *cascade*, Part IV).

What actually limits the original method: window or database? Exact ground truth



Mean excess over the optimum (ion trap, 3 qubits) by window w and database radius r. Red: the paper's $(8, 5)$.

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Mean excess over the optimum (ion trap, 3 qubits) by window w and database radius r. Red: the paper's $(8, 5)$.

- Larger window at $r = 5$: excess 46 $\rightarrow$ 43. Barely helps.
- Larger database at $w = 8$: excess 46 $\rightarrow$ 18.
- $w \geq 12$ and $r \geq 8$: excess ≤ 2.

Takeaway. The **database radius**, not the window or the search strategy, is the binding constraint. The theory predicts both are needed.

Four qubits: the journal benchmark

4q, 300 gates, 100 circuits	lower bound	CW-sem	CW-24	contiguous EW-16	port	published	Qiskit L3
ion trap: total	$\geq$11.1	**45.6**	56.7	110.3	78.8	111	164.9
IBM-style: total	$\geq$12.9	**49.3**	59.1	130.9	76.0	107	140.9
ion: RXX / IBM: CZ	–	–	27.6 / 27.1	44.2 / 47.1	33.3 / 32.4	43 / 43	47.3 / 47.3

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ion: RXX / IBM: CZ	–	–	27.6 / 27.1	44.2 / 47.1	33.3 / 32.4	43 / 43	47.3 / 47.3

- Same exact 3-qubit oracle, different windows: contiguous windows (EW-16) are *no better* than published; **DAG-convex windows** (CW-24) cut the published results by 49 % / 45 %; adaptive commutation-aware windows (CW-sem) go further.
- Two-qubit counts (27.6 RXX / 27.1 CZ) reach the journal's BQSKit numbers (28 / 37) without targeting them.
- No exact optimum at 4 qubits ($|G| = 1.2 \times 10^{13}$), but a certified lower bound; the remaining gap is at most $\approx 4\times$.

Part IV

Theory

Local rewriting as rewriting on a Cayley graph: when is it exact, what can it never do, and what do random benchmarks measure?

The model

- Alphabet Γ, group $G = \langle \Gamma \rangle$, word length $|g|$, ball $B(r)$, diameter D.
- Compute graph of depth d = ball $B(d)$ in $\mathrm{Cay}(G, \Gamma)$; stored words are geodesics.
- Rewriting system $\mathcal{R}(w, r)$: replace a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.

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- Rewriting system $\mathcal{R}(w, r)$: replace a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.
- Swapping commuting neighbours = moves in the **trace monoid**.
- Factors of trace-equivalent words = **convex** sets of the circuit DAG.
- So the paper's random commuting shuffles are a *stochastic enumeration of convex windows*; we enumerate them directly.
- The paper's V2 operates at $(w, r) = (8, 5)$.

Theorem 1: local = global above the diameter

Exactness threshold

Let G be finite with diameter D. If $w \geq D + 1$ and $r \geq D$, every (w, r)-reduced word is a geodesic.

Proof idea. A reduced word longer than D would have a first factor of length $D + 1$, which is not geodesic and fits in the window. So the word has length $\leq D < w$, and then the whole word is one window.

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Cascade

If at any time the current length $\ell \leq w$, the next sweep returns a geodesic. Relevant window: $w^\star = \min\{w : \ell(w) \leq w\}$.

- Measured $w^\star \approx 12$ (ion) / 16 (IBM-style) for 3 qubits, well below $D + 1 = 18$.
- The paper's $(8, 5)$ is below both thresholds.
- Continuous gate sets: $D = \infty$, no finite window is exact. Hence ground truth must come from the discrete case.

Theorem 2: gate-count parity

Parity

(a) For $n \leq 2$ the Cayley graph is **bipartite**: every equivalent rewrite removes an *even* number of gates.
(b) For $n \geq 3$, with any generating set containing square roots of each X_i, Z_i, it is **never** bipartite.

Proof idea. Bipartite $\iff$ a sign character χ with $\chi(\Gamma) = -1$. For $n \geq 3$, χ kills the Paulis and factors through $\mathrm{Sp}(2n, 2)$, which is simple: $\chi \equiv 1$.

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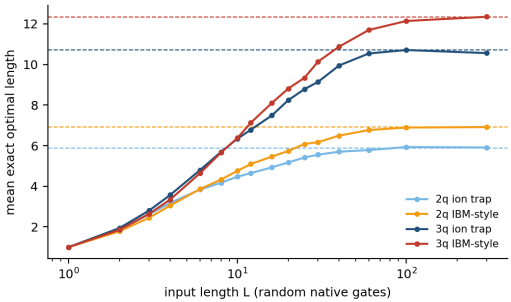
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- Exhaustive search, 3 qubits: the shortest odd relation has length $\rho = 11$ (ion) / 9 (IBM-style). Changing parity needs a window of ≥ 6 gates; smaller windows can never reach an optimum of the other parity.
- The paper's Fig. 4 shows a $6 \to 3$ replacement on two qubits: **impossible** with the stated gate set.

Benchmarks mix

A random word of length L is within $\frac{1}{2}\sqrt{|G|}\,\beta^L$ (total variation) of a uniform group element. So $\mathbb{E}[\text{optimal length}] \to$ the mean distance in G.

2 qubits, ion trap: $\beta = 0.937$ (sparse diagonalisation).

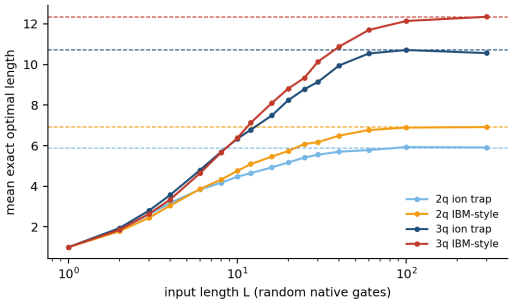


Theorem 3: random-circuit benchmarks measure a constant

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Takeaway. "Reduced from 300 to 45 gates" mostly measures the input length. Beyond $L \approx 100$, every random 3-qubit ion circuit has optimum ≈ 10.7 on average, whatever L is.

Lower bounds for every method (4 qubits)

Counting bound

$\Pr[|g| \leq d] \leq |B(d)|/|G|$, and spheres are submultiplicative, $|S(a+b)| \leq |S(a)||S(b)|$. Exact spheres up to a small depth bound $\mathbb{E}|g|$ from below for any optimiser.

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- Check at $n = 3$: bound 7.9 vs. truth 10.72.
- $n = 4$ (1.2×10^{13} elements), spheres to depth 6: $\mathbb{E}|g| \ge$ **11.1** (ion) / **12.9** (IBM-style).
- Best achieved: 45.6 / 49.3. Closing that gap is an open problem the framework makes precise.

Part V

A general optimiser

Drop the Clifford crutch: continuous angles, connectivity, calibrated noise, and a precise notion of “equivalent”.

Why go beyond Clifford?

The Clifford results don't transfer

- Real compilers emit **arbitrary angles** ($R_z(\theta)$, MS(θ)): no finite group, no tables.
- Real chips have **connectivity** (heavy-hex) and per-gate **calibrated errors**.
- "Up to global phase" is **unsafe** once a block is controlled.
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- Total gate count is the wrong objective when 2q gates dominate error.

Design decisions

- **No** Clifford assumption in the method; Clifford tables only to *evaluate*.
- Devices: IBM Heron (FakeTorino: $R_z(\theta)$, $\sqrt{X}$, X, CZ, heavy-hex, calibrated) and a trapped-ion model ($R_{x,y,z}(\theta)$, $R_{XX}(\theta)$, all-to-all).
- Semantics: exact incl. phase, state/measurement-aware, approximate with tracked error.
- Objective: chosen by noisy simulation, not assumed.

One idea: equivalence = group action, key = canonical orbit

notion	$U \sim V$ **iff**	**canonical database key**
exact	$V = U$	raw matrix
up to phase (the paper)	$V = e^{i\varphi}U$	phase-normalised matrix
input-aware (at start)	$V\|0\rangle = e^{i\varphi}U\|0\rangle$	phase-normalised $U\|0\rangle$
output-aware (before readout)	$V = DU$, D diagonal	rows phase-normalised
symmetry	$V = PUP^\dagger$, P a chip symmetry	minimum over the orbit
approximate	$\|U - V\| \leq \varepsilon$	exact key + verified distance

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One engine, many semantics

The database stores 64-bit hashes of canonical forms: one lookup finds the whole class. New semantics = new key function. Every hit is re-verified numerically.

Proposition 1 (state- and measurement-aware rewriting)

Program: prepare $|0^n\rangle$, run the circuit, measure all qubits. For a convex window S on wires W: (a) if nothing precedes S on W and $V|0_W\rangle = e^{i\varphi}U|0_W\rangle$, or (b) if nothing follows S on W and $V = DU$ with D diagonal, then replacing U by V leaves the output distribution unchanged, and so does any sequence of such rewrites.

Why. (a) the input is a product state; (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

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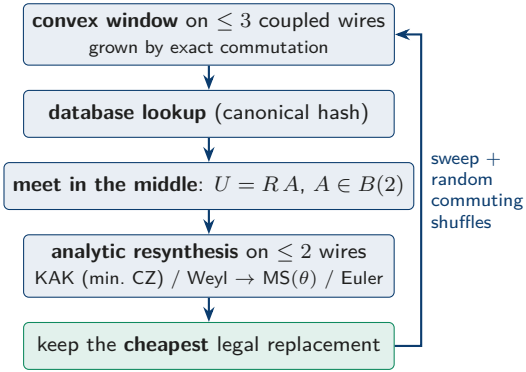
Why. (a) the input is a product state; (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

Proposition 2 (approximate rewriting)

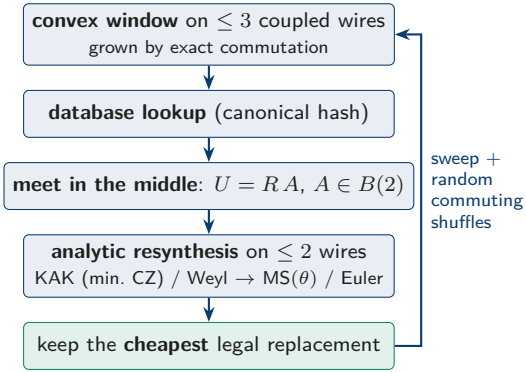
If rewrite i has operator-norm error ε_i, the final circuit is within $\sum_i \varepsilon_i$ of the original (including phase in exact mode).

Why. Telescoping, and $\|A(X-Y)B\| = \|X-Y\|$ for unitary A, B. The optimiser reports this bound with every circuit.

The engine


convex window on ≤ 3 coupled wires
grown by exact commutation
database lookup (canonical hash)
meet in the middle: $U = R\,A$, $A \in B(2)$
analytic resynthesis on ≤ 2 wires
KAN (min. CZ) / Weyl → MS(θ) / Euler
keep the **cheapest** legal replacement
sweep + random commuting shuffles

The engine



device, topology	notion	classes	depth
Heron, line	phase	6.31 M	9
	output-aware	0.34 M	10
	input-aware	1.41 M	14
ion, triangle	phase	28.4 M	6
	output-aware	7.36 M	6
	input-aware	31.6 M	7
paper ion alphabet	phase	2.93 M	9
paper IBM alphabet	phase	1.84 M	10

Grid: angles $\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi$; 23 (Heron) / 57 (ion) local gates. Build: 18 s – 38 min. Deepest level stored as keys only.

Global phase and cost: the details that matter

Exact including global phase

- Inside windows, match up to phase and record each phase.
- Native gates reach only a *discrete* set of phases: multiples of $\pi/4$ (Heron), $\pi/2$ (ion). Snap, or reject.
- Repair once at the end with the cheapest gadget; on ions absorb $\pm i$ into an existing MS gate: $R_{XX}(\theta \pm \pi)\, R_x(\pi)^{\otimes 2} = \pm i\, R_{XX}(\theta)$.
- Never return a costlier circuit than the input.

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Three candidate objectives

$$c_{\text{count}} = 1, \qquad c_{w2} = \begin{cases} 10 & \text{2q} \\ 1 & \text{1q} \end{cases}$$

$$c_{\text{cal}}(g) = -\log\bigl(1 - e(g)\bigr)$$

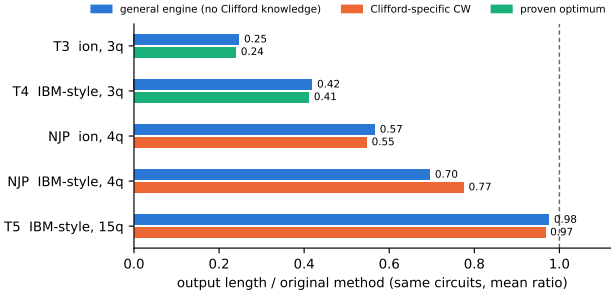
- $e^{-c_{\text{cal}}}$ = estimated success probability (ESP): minimising c_{cal} maximises ESP.
- $e(g)$ from FakeTorino calibration per qubit / pair; ion model $3 \cdot 10^{-5}$ / 10^{-3}; R_z virtual.
- Which one wins is decided by noisy simulation (Part VI).

Part VI

Results

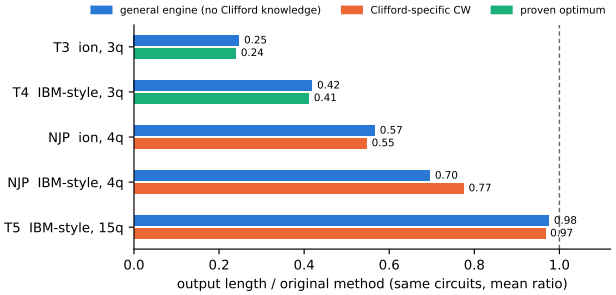
First the paper's own benchmarks against ground truth, then 60 standard circuits on hardware models.

Result 1: the paper's benchmarks, without Clifford knowledge



Output length / paper's method, same circuits and seeds. Paper semantics: up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets.

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Output length / paper's method, same circuits and seeds. Paper semantics: up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets.

- **3 qubits**: proven optimum on **17/20** (ion) and **18/20** (IBM-style); mean 10.8 vs. optimum 10.4.
- **4 qubits**: on par with the Clifford-specific optimiser (46.1 vs. 45.6; 51.4 vs. 49.3).
- **15 qubits**: all methods within 4 %.
- 3–9 s per circuit; all 250 outputs verified and native.

Takeaway. Knowing the group structure is not what makes the difference; large databases and convex windows are.

Result 2: standard suites on hardware models — setup

Circuits (60 per device, 2–12 qubits, $\leq$1,008 gates)

- Nam/VOQC arithmetic: Toffoli/Barenco ladders, adders, modular multiplication, $\mathrm{GF}(2^4)$ multiplier, product formulas, QFT8
- QASMBench small/medium (4 with classical feed-forward and 4 unparsable skipped)
- QFT, QAOA, Heisenberg Trotter, EfficientSU2, Steane encoder
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Pipelines (same native set, coupling map, seed)

- Qiskit transpile, levels 1 and 3
- TKET `FullPeepholeOptimise` on L1 + rebase
- ours on L1 output; **ours on L3 output**

Metrics: 2q count, total, depth, ESP, noisy-simulation TVD (≤ 8 active qubits); every output verified, native, coupled.

Result 2: relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
ours, approximate ($\varepsilon = 10^{-3}$), from L1	×1.01	**×0.84**	**×1.21**	**×0.95**	**×0.75**
ours, state/measurement-aware	**×0.77**	**×0.61**	**×1.28**	**×0.62**	**×0.50**

Ratios of summed counts; ESP: geometric mean of per-circuit ratios. Nam pf1_10/pf2_10 excluded: Qiskit L3 drops their 10^{-7} rad rotations (approximate, not exact).

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- **Stacked on L3**: never more 2q gates (Heron 9 better / 49 equal; ion 1 / 57) and never lower ESP, on either device.
- TKET keeps 2q counts close to L3 but needs 1.7–1.8× the gates after rebasing.
- Ion ESPs differ by $< 1\,\%$ across pipelines (MS error 10^{-3}), so they are omitted.

Ratios of summed counts; ESP: geometric mean of per-circuit ratios. Nam pf1_10/pf2_10 excluded: Qiskit L3 drops their 10^{-7} rad rotations (approximate, not exact).

Where do the gains come from? (honest accounting)

Stacked on L3, summed over 58 circuits

	Qiskit L3	+ ours
Heron $\sqrt{X}/X$ pulses	7,218	6,139
Heron R_z (virtual)	5,297	4,436
Heron CZ	2,889	2,830
ion R_x/R_y pulses	3,824	2,422
ion R_z (virtual)	2,223	**3,622**
ion MS	1,378	1,376

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- Heron: savings split between pulses and R_z; a few CZ.
- Ion: the calibrated objective *trades* pulses for free R_z. The total rises on 28/58 circuits, by design. "Never worse" holds for the objective, 2q count and ESP, **not** for the total.
- ESP $\times 1.21$ from L1 decomposes as $\times 1.11$ (Qiskit's L1 layout already beats its L3 layout) $\times$ $\times 1.09$ (our rewriting). We never change the layout.

Which objective? Let noisy simulation decide

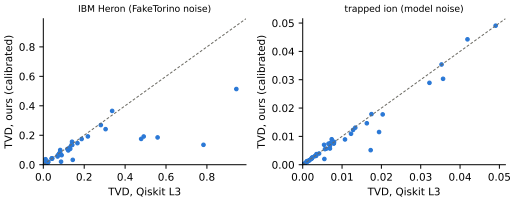
pipeline / objective	mean TVD (49 circuits) Heron	ion
Qiskit L1	0.115	0.0157
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ours, total count	0.096	0.0092
ours, 2q-weighted	**0.088**	**0.0084**
ours, calibrated $-\log F$	**0.088**	**0.0084**

Exact density-matrix probabilities (no shot noise). Heron: FakeTorino gate errors of the actual qubits. Ion: depolarising model.

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Exact density-matrix probabilities (no shot noise). Heron: FakeTorino gate errors of the actual qubits. Ion: depolarising model.



- Calibrated beats L3 on **39/49** (Heron) and 29/49 (ion) circuits.
- Calibrated $\approx$ 2q-weighted $>$ count $\Rightarrow$ calibrated is the default; 2q-weighted is a good proxy without calibration data.

The other semantics: state-aware and approximate

State/measurement-aware

- Largest gains: 2q gates $\times 0.77$ (Heron) / $\times 0.62$ (ion) relative to L3.
- Output distributions agree to $6 \cdot 10^{-12}$.
- Valid only for a complete program: $|0 \cdots 0\rangle$ in, full measurement out.

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Caveat: 5 (Heron) / 7 (ion) circuits are reversible logic on $|0 \cdots 0\rangle$ with a fixed basis-state output; they reduce to the empty circuit. Read this mode per circuit, not as an average.

Approximate ($\varepsilon = 10^{-3}$ per replacement)

- Almost no gain over exact on these suites: few near-identity blocks.
- Measured error always inside the tracked bound (max $6.4 \cdot 10^{-4}$).
- Exception: Nam pf2_10 (rotations $\sim 10^{-7}$ rad): 1,088 $\rightarrow$ 529 gates, error $1.3 \cdot 10^{-4} \leq$ bound $7.1 \cdot 10^{-4}$.

Verification: every number on these slides is checked

- **Exact mode**: $\|(U - V)\psi\|$ on random states, statevector on the active qubits: $\leq 8.2 \cdot 10^{-11}$.
- **State-aware**: TVD of output distributions: $\leq 6 \cdot 10^{-12}$.
- **Approximate**: measured error $\leq$ tracked bound, all runs.
- **Legality**: native gates only, 2q gates only on working couplers.

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- **Coverage**: 250/250 paper-benchmark outputs, 720/720 hardware-suite outputs.
- Nam pf2_10 on Heron is routed over 20 physical qubits, beyond the main run's 16-qubit cap: re-run and verified separately (same totals).
- FakeTorino has 11 dead couplers (error 1.0); removed for *all* pipelines after a first run put Qiskit L3 on them (ESP $\sim 10^{-22}$).

Part VII

Wrap-up

Lessons, limitations, critique of the original paper, and next steps.

Lesson: “exact” is easy to get subtly wrong

Found on Nam `pf1_10`/`pf2_10` (rotations of $\sim 10^{-7}$ rad): outputs first verified only to 10^{-6}.

Pitfalls

- `np.allclose` has a default *relative* tolerance 10^{-5}: near-commuting gates were swapped.
- $|\mathrm{tr}(U^\dagger V)|/d \approx 1$ is *quadratic* in the error: accepts $\|U - V\| \sim 10^{-4}$.
- Native gates reach only a *discrete* set of global phases.

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Fixes

- Linear operator-distance tests, one tolerance 10^{-10}, `rtol=0`.
- Replacement phases snapped to the reachable set; residuals rejected or charged to ε.
- Ion: $\pm i$ absorbed into an existing MS gate.

Takeaway. `pf1_10` now verifies to $5 \cdot 10^{-13}$ (was $1.8 \cdot 10^{-7}$). The original paper uses three different tolerances up to 10^{-3}.

Limitations of my work

- **Noise models, not hardware.** ESP and TVD from calibration snapshots; gate errors only (no readout, idle, crosstalk). Ion parameters are assumptions.
- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite; with phase as metadata (as Qiskit/TKET do), all gates $\times 0.838 \rightarrow \times 0.825$.

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- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite; with phase as metadata (as Qiskit/TKET do), all gates $\times 0.838 \rightarrow \times 0.825$.
- **Baselines.** No BQSKit (not installable here), no Quartz/QUESO; one seed for Qiskit and TKET.
- **Local only.** Layout and routing are Qiskit's; windows ≤ 3 qubits; single-threaded Python (median 10 s, max 180 s per circuit vs. $<$1 s for Qiskit).
- **Scope.** Circuits up to $\sim$1,000 gates; no mid-circuit measurement / feed-forward.

1. **Missing structure**: Clifford gate sets, but no tableaux, no exact keys; floating-point lookup with three tolerances; a classifier to speed up an avoidable $O(N)$ scan.
2. **No notion of optimality**: only compared with Qiskit; 4.5$\times$ from an optimum computable in minutes.
3. **Equivalence**: up-to-phase only, unsound under control; no state/measurement context; no error tracking.
4. **Hardware model**: discrete angles, all-to-all, total count. Matches neither IBM nor ion hardware.

Weaknesses of the original paper

1. **Missing structure**: Clifford gate sets, but no tableaux, no exact keys; floating-point lookup with three tolerances; a classifier to speed up an avoidable $O(N)$ scan.
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3. **Equivalence**: up-to-phase only, unsound under control; no state/measurement context; no error tracking.
4. **Hardware model**: discrete angles, all-to-all, total count. Matches neither IBM nor ion hardware.
5. **Evaluation**: random benchmarks mix, so "from 300 gates" measures input length; one circuit per table; MATLAB vs. compiled timings.
6. **Search**: contiguous windows, random swaps, no convergence criterion; radius 5 is the binding limit.
7. **V3**: unreported features/threshold; 30 % recall at default.
8. **Hardware evidence**: one 2-qubit circuit, no error bars; vanishes at Qiskit level 3.
9. **Reproducibility**: only V2 released, with visible bugs; database for Tables III–V unstated.

About the paper

- Reproduces, with qualifications on database, forest threshold, baselines and transpiler level.
- Its gates are Clifford: exact optima show it is 3–4.5$\times$ from optimal.
- Theory explains why: radius 5 and window 8 sit below the exactness threshold; random benchmarks measure a constant.

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- Reproduces, with qualifications on database, forest threshold, baselines and transpiler level.
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About the extension

- A general optimiser built on "equivalence = group action", with soundness and error bounds.
- Proven optimum on 35/40 of the paper's 3-qubit circuits without Clifford knowledge.
- On 58 real algorithms, stacked on Qiskit L3: never more 2q gates, never lower ESP, $-13\,\%$ gates on Heron; lowest noisy error with the calibrated objective.

Next steps, and questions for you

Next steps

1. Layout-aware rewriting with calibrated costs (the biggest remaining lever on Heron).
2. Cheaper exact phase repair on Heron.
3. 4-qubit windows via symmetry reduction, or a learned admissible A* heuristic trained on exact tables.
4. Ground truth beyond Clifford: exact Clifford+T normal forms for small T-counts.
5. BQSKit / Quartz baselines; real-hardware runs with error bars.

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Where I'd like your input

1. Is the target a short note on the reproduction + optimality gap, or a full paper on the general framework?
2. Is real-hardware access available for an error-barred experiment?
3. Which venue fits: quantum-computing (theory + benchmarks) or a compiler venue?
4. Should we contact the authors about the unstated database and the code bugs?

Deliverables

What	Where
Study notes (read instead of the paper)	`manuscripts/notes/notes.pdf`
Reproduction report (every table and figure)	`manuscripts/reproduction/reproduction_report.pdf`
Extension paper (12 pages)	`manuscripts/extension/extension_paper.pdf`
Python port, exact Clifford engine, general optimiser	`qcr/`
Experiment scripts and raw results	`experiments/`, `results/`
This talk	`manuscripts/talk/talk.pdf`

Thank you

Backup

Backup: paper-benchmark numbers (mean total gate count)

set	Qiskit L3	paper's method	Clifford CW-sem	optimum / bound	general
T3 (ion, 3q, 20)	127.2	47.1	–	10.4	**10.8**
T4 (IBM-style, 3q, 20)	104.0	36.8	–	12.2	**12.5**
journal ion (4q, 100)	164.9*	78.8*	45.6	$\geq$11.1	**46.1**
journal IBM-style (4q, 100)	140.9*	76.0*	49.3	$\geq$12.9	**51.4**
T5 (IBM-style, 15q, 10)	332.1	335.9	324.6	–	**327.7**

*first 30 circuits. General engine: up-to-phase semantics (as in the paper), total count, discrete $\pm\pi/2$ alphabets, all-to-all.

Backup: minimal entangling-gate counts, 3 qubits

entangling gates	0	1	2	3	4	5	6
3-qubit Cliffords	13,824	373,248	5,971,968	38,486,016	42,052,608	5,971,968	27,648

Mean 3.51, maximum 6. Identical for both alphabets ($R_{XX}(\frac{\pi}{2})$ and CZ are locally equivalent). For 2 qubits: 576 / 5,184 / 5,184 / 576 for 0–3, the known CNOT-count distribution.

Backup: worked examples from the reproduction

- **Fig. 1**: the 8-gate and 4-gate Bell circuits both equal CNOT $(H \otimes I)$ to 10^{-13}; no 3-gate circuit exists.
- **Fig. 4 analogue** (2 qubits, 40 gates): $40 \to 34 \to 26 \to 24 \to 20 \to 18 \to 16 \to 8 \to 4$; every step removes an even number of gates (parity theorem). The paper's final circuit contains an inverse pair; the optimum is 2.
- **Fig. 5 analogue** (3 qubits, 200 gates): $200 \to 36$ in 31 replacements.
- **Authors' demo circuit** (5 qubits, 300 gates): authors 140; our port 140 (132–148) with the demo database, 115 (112–117) with the 24-op depth-5 database.